College of Engineering – Department of Chemical Engineering

CURRICULUM

Bachelor of Science in Ceramic Engineering

Academic Year 2025-2026

Reference CMOs: CMO N0. 90 s. 2017, CMO No. 20, s. 2013 and CMO No. 39 s. 2021

Curriculum Description

Ceramic Engineering is the engineering discipline that studies the structure, properties, processing and applications of traditional and advanced inorganic and non-metallic materials, including bio mimic materials, that are generally processed at an elevated temperature into useful products. The application of Ceramic material is expanded tremendously due to the recent advances in the field of medicine which include bio-ceramic and other materials making ceramic engineering a booming field. The use of ceramic material is on the rise as it is a low-cost and efficient material. Ceramic engineering is needed right from the production of ceramic teeth, bones, and other fiber optic cables used for surgery to ceramic superconductors, lasers, etc. Ceramic engineering is also used in the field of construction, electronics, military, optical fibers, sports and transportation industry. The program will adopt outcome-based education (OBE) framework with flipped classroom and other blended learning pedagogies. Aside from lectures and laboratory activities, students are also given opportunities to do fieldworks, plant tours and site visits to reinforce the knowledge taught through experiential learning.

The program also aims to develop graduates with ethical behavior, integrity, social responsibility, effective communication, teamwork, and leadership skills, while promoting lifelong learning to keep up with the continuing technological advancement.

A professional Ceramic Engineer will assume leading roles in the technological and socio-economic development of both local and foreign industry in the field of ceramic materials, particularly in the areas of research, development, manufacturing, maintenance and operation. Graduates of this program are in academic, industrial, and government sectors. Job functions include research, teaching, process engineering and optimization and production of ceramic materials and related products.

Program Education Objectives

The ceramic engineering alumni three to five years after graduation shall:
PEO 1: Practice as specialist in solving complex ceramic engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Assume leadership position in industry, academe, government, or private sector with consideration to social and ethical responsibility.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes,

exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex engineering activities.

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
MATH 101 - Differential Calculus	4	0	4
MATH 102 - Integral Calculus	4	0	4
AI 102 – Probability and Statistics for AI	3	0	3
MATH 103 – Differential Equations	3	0	3
Subtotal	**14**	**0**	**14**
B. Natural/ Physical Sciences			
CHEM 101 – Chemistry for Engineers	3	3	4
PHYS 101 – Physics for Engineers	3	3	4
CHEM 103 – Soil Chemistry	3	0	3
Subtotal	**9**	**6**	**11**
C. Basic Engineering Sciences			
ENGG 101 – Introduction to Engineering	0	3	1
ENGG 102 – Engineering Drawing	0	3	1
ENGG 103 – Computer-Aided Design	0	6	2
ENGG 104 – Engineering Economics	3	0	3
ENGG 105 – Technopreneurship	3	0	3
ENGG 106 – Engineering Management	3	0	3
ENGG 113 – Environmental Science and Engineering	3	0	3
ENGG 120 – Mechanics of Rigid Bodies	4	0	4
ENGG 111 – Mechanics of Deformable Bodies	3	0	3

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ENGG 112 – Fluid Mechanics	3	0	3
ENGG 116 – Methods of Research	3	0	3
Subtotal	**25**	**12**	**29**
D. Allied Courses			
ChE 401 – Analytical Chemistry	4	3	5
ChE 403 – Ceramic Engineering for Calculation	2	3	3
CpE 101 – Computer Programming	0	3	1
EE 214 – Fundamentals of Electrical Engineering	3	0	3
GeoE 202 – Principles of Geology	3	0	3
CerE 307 – Elements of Mineralogy	3	0	3
CerE 203 – Fundamentals of Material Science and Engineering	3	0	3
Subtotal	**18**	**9**	**21**
E. Professional Courses			
CerE 406 – Advanced Ceramic	2	0	2
CerE 313 – Ceramic Manufacturing Technology	2	0	2
CerE 202 – Ceramic Engineering as Profession	2	0	1
CerE 205 – Ceramic Raw Materials and Processes	3	3	4
CerE 403 – Ceramic Engineering Design 1	1	3	2
CerE 409 – Ceramic Engineering Design 2	1	3	2
CerE 315 – Ceramic Engineering Project 1	1	3	2
CerE 408 – Ceramic Engineering Project 2	0	6	2
CerE 301 – Ceramic Measurement	0	6	2
CerE 308 – Computer Applications in Ceramic Engineering	0	6	2
CerE 201 – Crystal Chemistry	3	0	3
CerE 306 – Design and Analysis of Experiment in CerE	2	0	2
CerE 404 – Glass Manufacturing Technology	2	0	2
CerE 314 – Glazes and Enamels	2	3	3
CerE 410 – Integrative Studies	0	6	2
CerE 302 – Kinetics of Materials and Process	3	0	3
CerE 402 – Laws, Seminars, Safety and Plant Visits	2	3	3
CerE 401 – Microscopy and X-ray Characterization	2	0	2
CerE 303 – Phase and Equilibria in Ceramic Engineering	3	0	3
CerE 310 – Production of Ceramic Wares	3	6	5
CerE 312 – Properties of Ceramic	1	3	2
CerE 407 – Quality Assurance	2	0	1
CerE 309 – Refractories and Kiln Design	1	3	2
CerE 311 – Rheometry and Rheology	2	0	1
CerE 305 - Spectroscopy	2	0	2
CerE 204 – Thermodynamics of Materials	3	0	3
CerE 304 – Thermomechanical Properties	1	3	2
CerE 316 – CerE Elective 1 (Introduction to Heat and Mass Transfer)	2	0	2
CerE 405 – CerE Elective 2 (Introduction to Particle Technology)	2	0	2
ENGG 117 – OJT/Internship (320hrs)	0	320	4
Subtotal	**50**	**377**	**70**
TOTAL TECHNICAL COURSES	**116**	**404**	**145**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Ged 101 – Understanding the Self	3	0	3
Ged 102 – Mathematics in the Modern World	3	0	3
Ged 104 – The Contemporary World	3	0	3
Ged 105 – Readings in the Philippine History	3	0	3
Ged 106 – Purposive Communication	3	0	3
Ged 107 – Ethics	3	0	3
Ged 108 – Art Appreciation	3	0	3

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Ged 109 – Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives			
AI 101 – Linear Algebra for AI	3	0	3
AI 103 – Machine Learning and Neutral Networks	3	0	3
GEdE 101 – The Entrepreneurial Mind	3	0	3
GEd 103 – Life and Works of Rizal	3	0	3
Subtotal	**12**	**0**	**12**
C. Mandated Courses			
PATHFit 1 – Movement Competency Training	2	0	2
PATHFit 2 – Exercise-based Fitness Activities	2	0	2
PATHFit 3 – Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
PATHFit 4 - Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
NSTP 111 – National Service Training Program 1	3	0	3
NSTP 121 - National Service Training Program 2	3	0	3
Subtotal	**14**	**0**	**14**
TOTAL NON-TECHNICAL COURSES	**50**	**0**	**50**
GRAND TOTAL	**166**	**404**	**194**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**14**
B. Natural/Physical Sciences	**11**
C. Basic Engineering Sciences	**29**
D. Allied Courses	**21**
E. Professional Courses	**70**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives	**12**
C. Mandate Courses	**14**
GRAND TOTAL	**195**

CURRICULUM STRUCTURE

A. Curriculum Components

Specific Areas of the Program Curriculum			BatStateU-The NEU Revised Curriculum Structure
I. Technical Courses			
A. Mathematics			
1	MATH 101	Differential Calculus	4
2	MATH 102	Integral Calculus	4
3	MATH 103	Differential Equations	3
4	AI 102	Probability and Statistics for AI	3
		Sub-total	**14**
B. Natural/Physical Science			
1	CHEM 101	Chemistry for Engineers	4
2	PHYS 101	Physics for Engineers	4
3	CHEM 103	Soil Chemistry	3
		Sub-total	**11**
C. Basic Engineering Sciences			
1	ENGG 101	Introduction to Engineering	1
2	ENGG 102	Engineering Drawing	1
3	ENGG 103	Computer-Aided Design	2
4	ENGG 104	Engineering Economics	3
5	ENGG 105	Technopreneurship	3

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6	ENGG 106	Engineering Management	3
7	ENGG 113	Environmental Science and Engineering	3
8	ENGG 120	Mechanics of Rigid Bodies	4
9	ENGG 111	Mechanics of Deformable Bodies	3
10	ENGG 112	Fluid Mechanics	3
11	ENGG 116	Methods of Research	3
		Sub-total	**29**
D. Allied Courses			
1	ChE 401	Analytical Chemistry	5
2	ChE 403	Ceramic Engineering Calculation	3
3	CpE 101	Computer Programming	1
4	EE 214	Fundamentals of Electrical Engineering	3
5	GeoE 202	Principles of Geology	3
6	CerE 307	Elements of Mineralogy	3
7	CerE 203	Fundamentals of Material Science and Engineering	3
		Sub-total	**21**
E. Professional Engineering Courses			
1	CerE 406	Advanced Ceramic	2
2	CerE 313	Cement Manufacturing Technology	2
3	CerE 202	Ceramic Engineering as a Profession	1
4	CerE 205	Ceramic Raw Materials and Processes	4
5	CerE 403	Ceramic Engineering Design 1	2
6	CerE 409	Ceramic Engineering Design 2	2
7	CerE 315	Ceramic Engineering Project 1	2
8	CerE 408	Ceramic Engineering Project 2	2
9	CerE 301	Ceramic Measurement	2
10	CerE 308	Computer Applications in Ceramic Engineering	2
11	CerE 201	Crystal Chemistry	3
12	CerE 306	Design and Analysis of Experiment in CerE	2
13	CerE 404	Glass Manufacturing Technology	2
14	CerE 314	Glazes and Enamels	3
15	CerE 410	Integrative Studies	2
16	CerE 302	Kinetics of Materials and Process	3
17	CerE 402	Laws, Seminars, Safety and Plant Visits	3
18	CerE 401	Microscopy and X-ray Characterization	2
19	CerE 303	Phase Equilibria in Ceramic Engineering	3
20	CerE 310	Production of Ceramic Wares	5
21	CerE 312	Properties of Ceramic	2
22	CerE 407	Quality Assurance	1
23	CerE 309	Refractories and Kiln Design	2
24	CerE 311	Rheometry and Rheology	1
25	CerE 305	Spectroscopy	2
26	CerE 204	Thermodynamics of Materials	3
27	CerE 304	Thermomechanical Properties	2
		Sub-total	**62**
F. Elective Courses			
1	CerE 316	CerE Elective 1 (Introduction to Heat and Mass Transfer)	2
2	CerE 405	CerE Elective 2 (Introduction to Particle Technology)	2
		Sub-total	**4**
E. On-the-Job training			
1	ENGG 117	OJT/Internship (320 hrs)	4
		Sub-total	**4**
II. Non-Technical Courses			
A. Required General Education			
No.	**Course Code**	**Course Title**	**Units**
1	GEd 101	Understanding the Self	3
2	GEd 102	Mathematics in the Modern World	3
3	GEd 104	The Contemporary World	3
4	GEd 105	Readings in Philippine History	3
5	GEd 106	Purposive Communication	3

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6	GEd 107	Ethics	3
7	GEd 108	Art Appreciation	3
8	GEd 109	Science, Technology and Society	3
		Sub-total	**24**
B. General Education Elective and Mandated Courses			
1	AI 101	Linear Algebra for AI	3
2	AI 103	Machine Learning and Neural Networks	3
3	GEdE 101	The Entrepreneurial Mind	3
4	GEd 103	Life and Works of Rizal	3
		Sub-total	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)			
1	PATHFit 1	Movement Competency Training	2
2	PATHFit 2	Exercise-based Fitness Activities	2
3	PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2
4	PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2
		Sub-total	**8**
D. National Service Training Program			
1	NSTP 111	National Service Training Program 1	3
2	NSTP 121	National Service Training Program 2	3
		Sub-total	**6**
		Grand Total	**195**

SUMMARY

Specific Areas of the Program Curriculum	BatStateU-The NEU Revised Curriculum
I. Technical Courses	
A. Mathematics	**14**
B. Natural/Physical Science	**11**
C. Basic Engineering Sciences	**29**
D. Allied Courses	**21**
E. Professional Engineering Courses	**62**
F. Elective Courses	**4**
G. On-the-Job training	**4**
II. Non-Technical Courses	
A. Required General Education	**24**
B. General Education Elective and Mandated Courses	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)	**8**
D. National Service Training Program	**6**
Grand Total	**195**

A. Program of Study

BACHELOR OF SCIENCE IN CERAMIC ENGINEERING

CMO No. 90 series 2017, CMO No. 39, series of 2021, No. 27, series of 2015, No. 20, series of 2013

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 101	Introduction to Engineering	1	0	3			TC - C
GEd 104	The Contemporary World	3	3	0			NTC - A

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GEd 105	Readings in Philippine History	3	3	0			NTC - A
GEd 108	Art Appreciation	3	3	0			NTC - A
MATH 101	Differential Calculus	4	4	0			TC - A
NSTP 111	National Service Training Program 1	3	3	0			NTC - D
PATHFit 1	Movement Competency Training	2	2	0			NTC - C
CHEM 101	Chemistry for Engineers	4	3	3			TC - B
GEd 102	Mathematics in the Modern World	3	3	0			NTC - A
ENGG 102	Engineering Drawing	1	0	3			TC - C
	Total	**27**	**24**	**9**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CpE 101	Computer Programming	1	0	3			TC - C
GEd 101	Understanding the Self	3	3	0			NTC - A
GEd 106	Purposive Communication	3	3	0			NTC - A
GEd 109	Science, Technology and Society	3	3	0	ENGG 401		NTC - A
MATH 102	Integral Calculus	4	4	0	MATH 101		TC - A
PHYS 101	Physics for Engineers	4	3	3	MATH 101		TC - B
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111		NTC - D
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1		NTC - C
GEdE 101	The Entrepreneurial mind	3	3	0			NTC - B
	Total	**26**	**24**	**6**			

FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GEd 103	Life and Works of Rizal	3	3	0			NTC - B
CHEM 103	Soil Chemistry	3	3	0	CHEM 101		TC-B
GEd 107	Ethics	3	3	0			NTC - A
	Total	**9**	**9**	**0**			

SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ChE 401	Analytical Chemistry	5	4	3	CHEM 101		TC-D
ChE 403	Ceramic Engineering Calculation	3	2	3	CHEM 101		TC-D
CerE 201	Crystal Chemistry	3	3	0	CHEM 101		TC-E
ENGG 120	Mechanics of Rigid Bodies	4	4	0	PHYS 101, MATH 102		TC-C
MATH 103	Differential Equations	3	3	0	MATH 102		TC-A
CerE 202	Ceramic Engineering as a Profession	1	2	0			TC-E
CerE 203	Fundamentals of Material Science and Engineering	3	3	0	CHEM 101, PHYS 101		TC-C
AI 101	Linear Algebra for AI	3	3	0	MATH 102		NTC-B

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PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**27**	**26**	**6**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GeoE 202	Principles of Geology	3	3	0	CHEM 103		TC-B
CerE 204	Thermodynamics of Materials	3	3	0	CHEM 101, PHYS 101		TC-E
ENGG 111	Mechanics of Deformable Bodies	3	3	0	PHYS 101, MATH 102		TC-C
ENGG 112	Fluid Mechanics	3	3	0	ChE 403, MATH 103		TC-D
CerE 205	Ceramic Raw Materials and Processes	4	3	3	ChE 401, CerE 203		TC-E
ENGG 103	Computer-Aided Design	2	0	6	ENGG 102		TC-C
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		TC-A
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		NTC-B
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2		NTC-C
	Total	**26**	**23**	**9**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CerE 301	Ceramic Measurement	2	0	6			TC-E
CerE 302	Kinetics of Materials and Process	3	3	0	CerE 204		TC-E
CerE 303	Phase Equilibria in Ceramic Engineering	3	3	0	CerE 204		TC-E
CerE 304	Thermomechanical Properties	2	1	3	CerE 203		TC-E
CerE 305	Spectroscopy	2	2	0	ChE 401		TC-E
CerE 306	Design and Analysis of Experiment in CerE	2	2	0	AI 102		TC-E
ENGG 116	Methods of Research	3	3	0	3rd Year Standing, AI 102		TC-E
AI 103	Machine Learning and Neural Networks	3	2	3	GEdE 101, AI 102, CpE 101		NTC-B
CerE 307	Elements of Mineralogy	3	3	0	GeoE 202, CHEM 103		TC-E
CerE 308	Computer Applications in Ceramic Engineering	2	0	6	MATH 103, CpE 101		TC-E
	Total	**25**	**19**	**18**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CerE 309	Refractories and Kiln Design	2	1	3	GeoE 202, CerE 205		TC-E
CerE 310	Production of Ceramic Wares	5	3	6	CerE 204, CerE 205		TC-E

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CerE 311	Rheometry and Rheology	1	2	0	ChE 401		TC-E
CerE 312	Properties of Ceramic	2	1	3	CerE 204, CerE 205		TC-E
CerE 313	Cement Manufacturing Technology	2	2	0	CerE 205		TC-E
CerE 314	Glazes and Enamels	3	2	3	CerE 303, CerE 304		TC-E
ENGG 105	Technopreneurship	3	3	0	Third Year Standing		TC-C
EE 214	Fundamentals of Electrical Engineering	3	3	0	PHYS 101		TC-D
CerE 315	Ceramic Engineering Project 1	2	1	3	ENGG 116, CerE 306		TC-E
	Total	**23**	**18**	**18**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CerE 316	CerE Elective 1 (Introduction to Heat and Mass Transfer)	2	2	0			TC-E
ENGG 104	Engineering Economics	3	3	0	Third Year Standing, AI 102		TC-C
ENGG 106	Engineering Management	3	3	0	Third Year Standing		TC-C
	Total	**8**	**8**	**0**			

FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
CerE 401	Microscopy and X-ray Characterization	2	2	0	CerE 307		TC-E
CerE 402	Laws, Seminars, Safety and Plant Visits	3	2	3	Fourth Year Standing		TC-E
CerE 403	Ceramic Engineering Design 1	2	1	3	Fourth Year Standing		TC-E
CerE 404	Glass Manufacturing Technology	2	2	0	CerE 205		TC-E
CerE 405	CerE Elective 2 (Introduction to Particle Technology)	2	2	0			TC-E
CerE 406	Advanced Ceramic	2	2	0	CerE 201, CerE 205		TC-E
CerE 407	Quality Assurance	1	2	0	AI 102		TC-E
CerE 408	Ceramic Engineering Project 2	2	0	6	CerE 315		TC-E
	Total	**16**	**13**	**12**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 117	OJT/Internship (320 hrs)	4		320 hours	Fourth Year Standing		TC-F
CerE 409	Ceramic Engineering Design 2	2	1	3	CerE 403, ENGG 106		TC-E
CerE 410	Integrative Studies	2	0	6	Graduating		TC-E
	Total	**8**	**1**	**9**			

TOTAL UNITS: 195